Lemma: Let G be a group $|G| = p^n$ -elements.

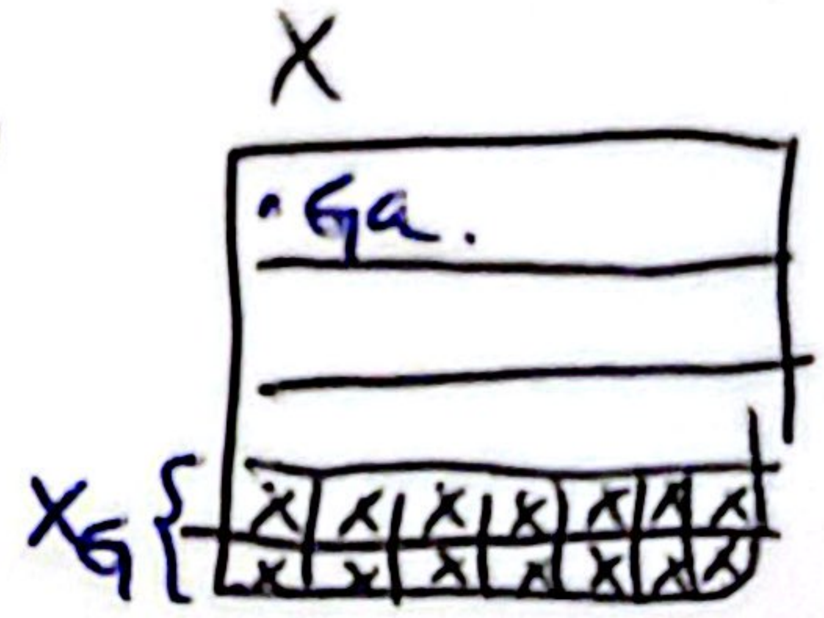
$G \curvearrowright X$ and $X_G = \{x \in X \mid \forall g \in G \mid g \cdot x = x\}$

then $|X| \equiv_p |X_G|$

Proof: let $a_1, \ldots, a_r$ be a set of representatives in X of $\sim_G$

then $X = \dot{\bigcup}_{i=1}^{r} G \cdot a_i = \{a_1\} \cup \ldots \cup \{a_k\} + \dot{\bigcup}_{i=1}^{s} G a_i'$

(orbits of size 1) + other orbits.



~~$G \cdot a_i = \{g \cdot a \mid g \in G\}$~~

$G \cdot a_i = \{g \cdot a \mid g \in G\}$

$\Rightarrow |X| = \sum_{i=1}^{r} |G a_i| = \underbrace{1 + \cdots + 1}_{|X_G|\text{-times}} + \sum_{i=1}^{s} |G a_i'|$

$\boxed{\text{Claim}}$ $p \mid |G \cdot a_i'|$

$\boxed{\text{Problem for the HWK:}}$ let G be a group acting on X.

then $|G \cdot a| = (G : G_a)$ where

$G_a = \{g \in G \mid g a = a\}$

Then By hypotesis $|G \cdot a| \geq 2$ since $|G| = p^n$ by lagrange

$|G a| = (G : G_a) = p^k$ then $p \mid |G \cdot a_i'|$.

thus $|X| \equiv_p |X_G|$

The Sylow Theorems:

Context: Let G be a group and $\mathcal{S} = \{S \subseteq G \mid S \leq G\}$

we consider the action of G on $\mathcal{S}$

$$\begin{aligned} \psi : G \times \mathcal{S} &\longrightarrow \mathcal{S} \\ (g, S) &\longrightarrow gSg^{-1} \end{aligned}$$

Given $H \leq G$ fixed we define the isotropy subgroup of H

$$G_H = \{ g \in G \mid gHg^{-1} = H \}$$

and $G_H \leq G$. this is the group of elements that leave H invariant under conjugation.

We also refer to G_H as the NORMALIZER OF H in G

~~sub~~ $\boxed{\text{Lemma}}$ G_H is a subgroup:

a). $e_G \in G_H$ $\quad e_G H e_G^{-1} = H$.

b). let $g, l \in G_H \Rightarrow gHg^{-1} = H$ and $lHl^{-1} = H$

$$glH(gl)^{-1} = g\underbrace{lHl^{-1}}_{=H}g^{-1} = gHg^{-1} = H.$$

c). <u>inverse</u>: $g \in G_H$ $\quad gHg^{-1} = H \Leftrightarrow H = g^{-1}Hg$.

<u>Remark</u>: If H is finite $H \leq G$ $\quad gHg^{-1} = H \Leftrightarrow \forall h \in H$ $ghg^{-1} \in H$.

<u>Why?</u> Conjugation is injective: $\begin{aligned} G &\longrightarrow G \\ h &\longrightarrow ghg^{-1} \end{aligned}$

$\Rightarrow h_1, h_2 \in G$ $\quad gh_1g^{-1} = gh_2g^{-1} \Rightarrow h_1 = h_2$.

thus for $g \in N[H]$ $\quad i_g : H \to H$

$h \to ghg^{-1}$ is injective.

H is finite $\Rightarrow$ any injective map is a bijection so $gHg^{-1} = H$.

for $g \in N[H]$.

<u>Recall:</u>

<u>Definition:</u> $(G, \circ_G, e_G)$ a finite group is said to be a p-group if ~~every element has order p~~. if $|G| = p^n$.

GOAL 1:

$\boxed{\text{Theorem}}$: (First Syllow theorem) Let G be a finite group and let $|G| = p^n \cdot m$ for $n \geq 1$ and where $p \nmid m$.

<u>Then:</u>

1. G contains a subgroup of order p^i for each $1 \leq i \leq n$.
2. Every subgroup H of G of order p^i is a normal subgroup of a subgroup of order p^{i+1} for $1 \leq i \lneq n$.

For this we need:

- $\boxed{\text{Cauchy's theorem}}$: Let G be a group such that $p \mid |G|$ then G has a subgroup of order p.
- <u>Lemma:</u> H a p-subgroup of a finite group G. Then
$$(N[H] : H) \equiv_p (G : H)$$

<u>Lemma:</u> $G \curvearrowright X$ $\quad |G| = p^n$ let $X_G = \{x \in X \mid g \cdot x = x\}$

$\Rightarrow$ ~~[illegible]~~ $|X| \equiv_p |X_G|$ $\qquad G \cdot a = \{g \cdot a \mid g \in G\}$

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Cauchy's theorem: Let G be a group such that $p \mid |G|$. Then G has an element of order p and therefore a subgroup of order p.

Proof: Let

$$X = \{ \underbrace{(g_0, \dots, g_{p-1})}_{p\text{-tuples}} \mid g_0, \dots, g_{p-1} \in G \quad \underbrace{g_0 g_1 \cdots g_{p-1} = e_G}_{\text{multiplied together are the identity}} \}$$

i.e.

Since e_G: $g_0 = (g_1 \cdots g_{p-1})^{-1}$

Thus given $g_1, \dots, g_{p-1}$ the eq holds taking $g_0 = (g_1 \cdots g_{p-1})^{-1}$

Thus $|X| = |G|^{p-1}$. $\boxed{\text{Since } p \mid |G| \Rightarrow p \mid |X|}$

Suppose that $(g_0, \dots, g_{p-1}) \in X \Rightarrow g_0 = (g_1 \cdots g_{p-1})^{-1}$

$\Rightarrow (g_1, \dots, g_{p-1}, g_0) \in X$

Repeating this process $g_1 = (g_2 g_3 \cdots g_{p-1} g_0)^{-1}$ then

$(g_2, \dots, g_{p-1}, g_0, g_1) \in X$

Then for any $k \in \mathbb{Z}_p$

$$(g_k, g_{k+_p 1}, g_{k+_p 2}, \dots, g_{k+_p (p-1)}) \in X.$$

Then we have an action of $\mathbb{Z}_p$ on X:

Let $k \in \mathbb{Z}_p$ and $(g_0,\dots,g_{p-1}) \in X$ then

$$k(g_0,\dots,g_{p-1}) = (g_k, g_{k+_p 1}, \dots, g_{k+_p(p-1)}) \in X$$

This is an action since:

$$0(g_0,\dots,g_{p-1}) = (g_0, g_1, \dots, g_{p-1})$$

$$k(\ell(g_0,\dots,g_{p-1})) = k(g_\ell, g_{\ell+_p 1}, \dots, g_{\ell+_p(p-1)})$$

$$= (g_{k+_p\ell}, g_{k+_p\ell+_p 1}, \dots, g_{k+_p\ell+_p(p-1)}) \in X$$

$$= (k+_p\ell)(g_0,\dots,g_{p-1})$$

Then we have an action $\mathbb{Z}_p \curvearrowright X$

let $X_{\mathbb{Z}_p}$ be the elements of X of orbit 1.

Recall

$G \curvearrowright X$ an ~~action~~ orbit for $x_0 \in X$

$G \cdot x_0 = \{g x_0 \mid g \in G\}$

$\boxed{\text{Claim}}$ ~~Then~~ $|X| \equiv_p |X_{\mathbb{Z}_p}|$

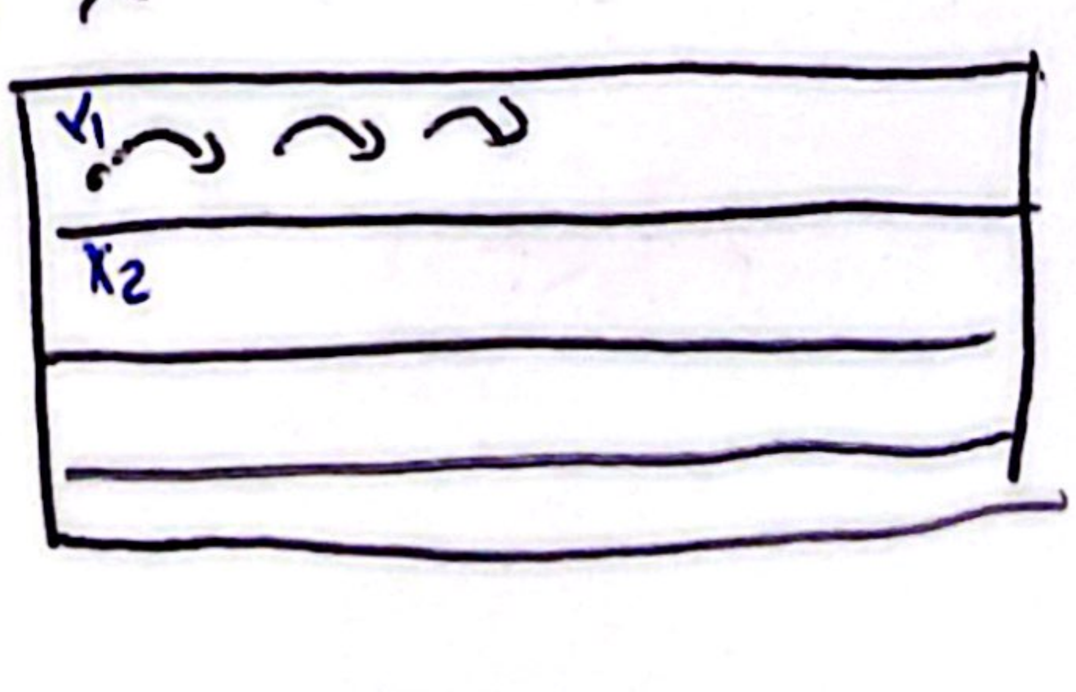
Proof of claim:

$r \rightarrow$ # of orbits.

$$X = \sum_{i=1}^{r} |\mathbb{Z}_p \cdot x_i|$$

Pick an element on each orbit

X



then $|X| = \underbrace{|X_{\mathbb{Z}_p}|}_{\text{orbit of size 1.}} + \sum_{i=1}^{s} \underbrace{|\mathbb{Z}_p \cdot x_i|}_{\text{sum of other orbits of size} > 1}$ (2)

To show that $|X| \equiv_p |X_{\mathbb{Z}_p}|$ we must argue that $p \mid |\mathbb{Z}_p \cdot x_i|$ (for those in (2))

$|\mathbb{Z}_p \cdot x_i| \geq 2$

$\boxed{\text{HWK}}$: Let G be a group and $x \in X \leftarrow$ a set. Assume $G \curvearrowright X$ is an action then $|G \cdot x| = (G : G_x)$ $\quad G_x = \{g \in G \mid gx = x\}$ $\to$ stabalizer of x.

Since $|\mathbb{Z}_p \cdot x_i| = (\cancel{G}\, \mathbb{Z}_p : \mathbb{Z}_{p_x}) = p$.

then $|\mathbb{Z}_p \cdot x_i| = p$.

thus $|X| \equiv_p |X_{\mathbb{Z}_p}|$.

Thus we have $|X| \equiv_p |X_{\mathbb{Z}_p}|$ then the p-tuple $(e, \ldots, e) \in X_{\mathbb{Z}_p}$ since re-arranging the entries does not change the p-tuple. End of claim.

Since $p \mid |X|$ and $|X| \equiv_p |X_{\mathbb{Z}_p}|$ then $|X_{\mathbb{Z}_p}| \equiv_p 0$.

Since $|X_{\mathbb{Z}_p}| \geq 1$ as $(e, \ldots, e) \in X_{\mathbb{Z}_p}$ $\Rightarrow \exists$ a tuple $(a, \ldots, a) \in X_{\mathbb{Z}_p}$ with $a \neq e$ and $a^p = e$.

so a has order p and $H = \langle a \rangle$ is the subgroup of order p of G.

Lemma: Let H be a p-group of a finite group G.

then $(N[H]:H) \equiv_p (G:H)$

Proof: sea $\mathcal{L} = \{gH \mid g \in G\} \to$ left cosets.

And we can consider H acting on $\mathcal{L}$ by left translation

so that $h \cdot (gH) = (hg)H$. $\quad H \curvearrowright \mathcal{L}_H$.

Let $\mathcal{L}_H = \{gH \in \mathcal{L} \mid \forall h \in H \quad hgH = gH\}$ $\leftarrow$ left coset invariant under the action of H.

$hgH = gH \iff g^{-1}hgH = H \iff g^{-1}hg \in H$.

$\iff g^{-1}Hg = H \iff g^{-1} \in N[H]$.

$\iff g \in N[H]$.

Then $gH \in \mathcal{L}_H \iff g \in N[H]$

Thus the left cosets in $\mathcal{L}_H$ are those contained in $N[H]$

# of cosets is $(N[H]:H) \Rightarrow |\mathcal{L}_H| = (N[H]:H)$

Since H is a p-group $|H| = p^k$ for some k

$\Rightarrow |\mathcal{L}| \equiv |\mathcal{L}_H| \mod p$. i.e. $(G:H) \equiv_p (N[H]:H)$

Thm: Let G be a finite group and $|G| = p^n \cdot m$ where $n \geq 1$ and $p \nmid m$. Then:

1) G contains a subgroup of order p^i $\forall i$ $1 \leq i \leq n$.
2) Every subgroup H of G of order p^i is a normal subgroup of ~~order p~~ a subgroup of order p^{i+1} $1 \leq i < n$.

Proof:

① By Cauchy's theorem G contains a subgroup of order p.
$\boxed{\text{By induction we show}}$ the existence of a subgroup of order p^i for $i < n$. implies the existence of a subgroup of order p^{i+1}.

Proof of the inductive step: Let H be a subgroup of order p^i. Since $i < n$ then $p \mid (G:H)$ By the lemma $p \mid (N[H]:H)$ since H is a normal subgroup of $N[H]$ we can form the quotient $N[H]/H$ then $p \mid |N[H]/H|$

By Cauchy's theorem the factor group $N[H]/H$ has a subgroup K of order p.

let $\pi : N[H] \to N[H]/H$
$n \longrightarrow nH$

Then $\pi^{-1}(K)$ is a subgroup of $N[H]$ that contains H, hence it is a subgroup of G.

and its size es p^{i+1}.

(2) every subgroup H of G of order p^i is a normal subgroup of a group of order p^{i+1}.

This is given by the previous construction, given H we find $H < \pi^{-1}(K) \leq N[H]$ where $|\pi^{-1}[K]| = p^{i+1}$. Since $H \trianglelefteq N[H] \Rightarrow H \trianglelefteq \pi^{-1}[K]$.

<u>Definition</u>: A <u>Sylow p-subgroup</u> P of a group G <u>is a maximal p-subgroup of G</u> that is: a p-subgroup contained in no larger p-subgroup.

Let G be a finite group and assume $|G| = p^n \cdot m$ with $p \nmid m$. the previous theorem shows that there is a subgroup of G of size p^n, let's name it P.

then P is a Syllow-p-group.

Note that P is a Syllow p-group then any conjugate is also a Syllow p-group.

$\Downarrow$

turns out they all come of this form.

⑤

(Second Syllow's theorem) Let P_1 and P_2 be Syllow p-subgroups of a finite group G. Then P_1 and P_2 are conjugate subgroups of G.

<u>Proof:</u> We will let one of the subgroups act on left cosets of the other one and use theorem 14.19.

Let $\mathcal{L} = \{gP_1 \mid g \in G\} \to$ left cosets of P_1

And let $P_2 \curvearrowright \mathcal{L}$

$$\mu : P_2 \times \mathcal{L} \to \mathcal{L}$$
$$(h, gP_1) \to hgP_1$$

By the lemma $|\mathcal{L}| \equiv_p |\mathcal{L}_{P_2}|$ where

$\mathcal{L}_{P_2} = \{gP_1 \in \mathcal{L} \mid \forall h \in P_2 \ \ h\cdot(gP_1) = gP_1\}$.

Since P_1 is a p-Syllow group $(G:P_1) = |\mathcal{L}|$ is not div by p.

so $|\mathcal{L}_{P_2}| \neq 0$. Let $xP_1 \in \mathcal{L}_{P_2} \Rightarrow yxP_1 = xP_1 \ \ \forall y \in P_2$.

so $x^{-1}yxP_1 = P_1 \ \ \forall y \in P_2$.

$\Rightarrow x^{-1}yx \in P_1 \ \ \forall y \in P_2$

$\Rightarrow x^{-1}P_2x \leq P_1 \ \ \Rightarrow (P_1 : x^{-1}P_2x) = p^k$

since P_2 is max $\Rightarrow k = 0$.

$\Rightarrow P_1 = x^{-1}P_2x$. ☒

<u>Third Syllow theorem</u>: If G is a finite group and $p \mid |G| \Rightarrow$ # of Sylow p-subgroup is congruent to 1 mod p and divides $|G|$.

<u>Proof</u>: Let P be one Sylow p-subgroup of G.

Let $\mathcal{L} = \{ P_0 \leq G \mid P_0 \text{ is a Syllow } p\text{-group} \}$

And let P act on $\mathcal{L}$ by conjugation.

$$P \times \mathcal{L} \longrightarrow \mathcal{L}$$
$$(x, T) \longrightarrow xTx^{-1}$$

By theorem 14.19. $|\mathcal{L}| \equiv_{\text{mod } p} |\mathcal{L}_P|$

$\mathcal{L}_P = \{ T \mid \forall x \in P \ \ xT = T \}$.

If $T \in \mathcal{L}_P$ then $xTx^{-1} = T \ \ \forall x \in P$. thus.

$P \leq N[T]$. since $T \leq N[T]$ and both P and T are Sylow p-subgroup of G, $\Rightarrow$ they are also P-sub. of $N[T]$. $\underset{\text{2nd theorem}}{\Rightarrow}$ then they are conjugate in $N[T]$.

Since $T \trianglelefteq N[T] \Rightarrow T$ is its only conjugate in $N[T]$

$\Rightarrow T = P$. then $\mathcal{L}_P = \{P\}$

since $|\mathcal{L}| \equiv_{\text{mod } p} |\mathcal{L}_P| = 1 \Rightarrow$ # of Syllow p-groups is $\equiv_p 1$.

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Now let G act on $\mathcal{L}$ by conjugation. Since all Sylow p-subgroups are conjugate there is only one orbit in $\mathcal{L}$ under G.

If $P \in \mathcal{L}$ then $|\mathcal{L}| = |\text{orbit of } P| = (G : G_P)$

(G_P is the normalizer of P)

Since $(G : G_P) \mid |G| \Rightarrow$ the # of p-syllow groups divides $|G|$